WEEK 2 - RELATIONAL SCHEMA AND MYSQL DATABASE

Campus Equipment Borrowing and Inventory Management System

Class Code: 4-311

1. Relational Schema

The design uses the same six entities as the Week 1 ERD. PK indicates a primary key; FK a foreign key; UQ a unique constraint.

```
USER_ACCOUNT(
    user_id PK,
    university_id UQ,
    full_name,
    email UQ,
    role,
    is_active
)
```

```
CATEGORY(
    category_id PK,
    category_name UQ,
    description
)
```

```
EQUIPMENT_ITEM(
    item_id PK,
    category_id FK,
    asset_tag UQ,
    item_name,
    location,
    condition_status,
    lifecycle_status
)
```

```
BORROW_TRANSACTION(
    transaction_id PK,
    borrower_id FK,
    processed_by FK (nullable),
    requested_at,
    requested_due_at,
    released_at (nullable),
    due_at (nullable),
    status
)
```

```
BORROW_ITEM(
    transaction_id PK, FK,
    item_id PK, FK,
    condition_out (nullable),
    returned_at (nullable),
    condition_in (nullable)
)
```

```
MAINTENANCE_RECORD(
    record_id PK,
    item_id FK,
    reported_by FK,
    reported_at,
    issue_description,
    status,
    resolved_at (nullable)
)
```

2. Normalization

First Normal Form (1NF). Each column holds one atomic value, with no repeating groups. A transaction with multiple equipment units is represented by separate BORROW_ITEM rows rather than item_1, item_2, and similar columns.

Second Normal Form (2NF). All tables satisfy 1NF. Five tables have a single-column primary key. BORROW_ITEM has the composite key (transaction_id, item_id); its condition and return attributes describe that specific item within that specific transaction, so there is no partial dependency.

Third Normal Form (3NF). All non-key attributes depend on the key and not on other non-key attributes. Borrower details are stored in USER_ACCOUNT, category descriptions in CATEGORY, and equipment details in EQUIPMENT_ITEM. BORROW_TRANSACTION, BORROW_ITEM, and MAINTENANCE_RECORD reference these records using foreign keys instead of duplicating them.

Result: All six tables are designed to satisfy 3NF, reducing update anomalies and unnecessary duplication.

3. Data Dictionary

Data types and key definitions below match the accompanying MySQL script. PK = primary key; FK = foreign key; UQ = unique; - = non-key attribute. Optional fields are identified in their descriptions or schema.

Table	Attribute	Data Type	Key	Description
USER_ACCOUNT	user_id	INT UNSIGNED	PK	Unique system user identifier
USER_ACCOUNT	university_id	VARCHAR(20)	UQ	Unique university or employee ID
USER_ACCOUNT	full_name	VARCHAR(100)	-	Name of the account holder
USER_ACCOUNT	email	VARCHAR(120)	UQ	Unique account email address
USER_ACCOUNT	role	ENUM	-	STUDENT, STAFF, or ADMIN
USER_ACCOUNT	is_active	BOOLEAN	-	Whether the account is active
CATEGORY	category_id	INT UNSIGNED	PK	Unique category identifier
CATEGORY	category_name	VARCHAR(80)	UQ	Unique equipment category name
CATEGORY	description	VARCHAR(255)	-	Optional category description
EQUIPMENT_ITEM	item_id	INT UNSIGNED	PK	Unique equipment unit identifier
EQUIPMENT_ITEM	category_id	INT UNSIGNED	FK	Category for this equipment unit
EQUIPMENT_ITEM	asset_tag	VARCHAR(30)	UQ	Unique physical asset tag
EQUIPMENT_ITEM	item_name	VARCHAR(120)	-	Equipment model or descriptive name
EQUIPMENT_ITEM	location	VARCHAR(100)	-	Usual storage or assigned location
EQUIPMENT_ITEM	condition_status	ENUM	-	GOOD, FAIR, DAMAGED, UNDER_REPAIR
EQUIPMENT_ITEM	lifecycle_status	ENUM	-	ACTIVE, MAINTENANCE, RETIRED
BORROW_TRANSACTION	transaction_id	INT UNSIGNED	PK	Unique borrowing request/transaction
BORROW_TRANSACTION	borrower_id	INT UNSIGNED	FK	User requesting the equipment
BORROW_TRANSACTION	processed_by	INT UNSIGNED	FK	Optional staff/admin reviewer
BORROW_TRANSACTION	requested_at	DATETIME	-	Date and time request was entered
BORROW_TRANSACTION	requested_due_at	DATETIME	-	Requested return deadline

Table	Attribute	Data Type	Key	Description
BORROW_TRANSACTION	released_at	DATETIME	-	Actual equipment release time; optional
BORROW_TRANSACTION	due_at	DATETIME	-	Confirmed return deadline; optional
BORROW_TRANSACTION	status	ENUM	-	PENDING, APPROVED, REJECTED, CANCELLED, RELEASED, CLOSED
BORROW_ITEM	transaction_id	INT UNSIGNED	PK, FK	Borrowing transaction containing item
BORROW_ITEM	item_id	INT UNSIGNED	PK, FK	Specific equipment unit borrowed
BORROW_ITEM	condition_out	ENUM	-	Condition at release; may be empty
BORROW_ITEM	returned_at	DATETIME	-	Actual return time, if returned
BORROW_ITEM	condition_in	ENUM	-	Condition upon return, if returned
MAINTENANCE_RECORD	record_id	INT UNSIGNED	PK	Unique maintenance issue record
MAINTENANCE_RECORD	item_id	INT UNSIGNED	FK	Equipment item needing maintenance
MAINTENANCE_RECORD	reported_by	INT UNSIGNED	FK	User who reported the issue
MAINTENANCE_RECORD	reported_at	DATETIME	-	Date and time issue was reported
MAINTENANCE_RECORD	issue_description	TEXT	-	Description of the observed issue
MAINTENANCE_RECORD	status	ENUM	-	OPEN, IN_PROGRESS, or RESOLVED
MAINTENANCE_RECORD	resolved_at	DATETIME	-	Issue resolution time, if completed

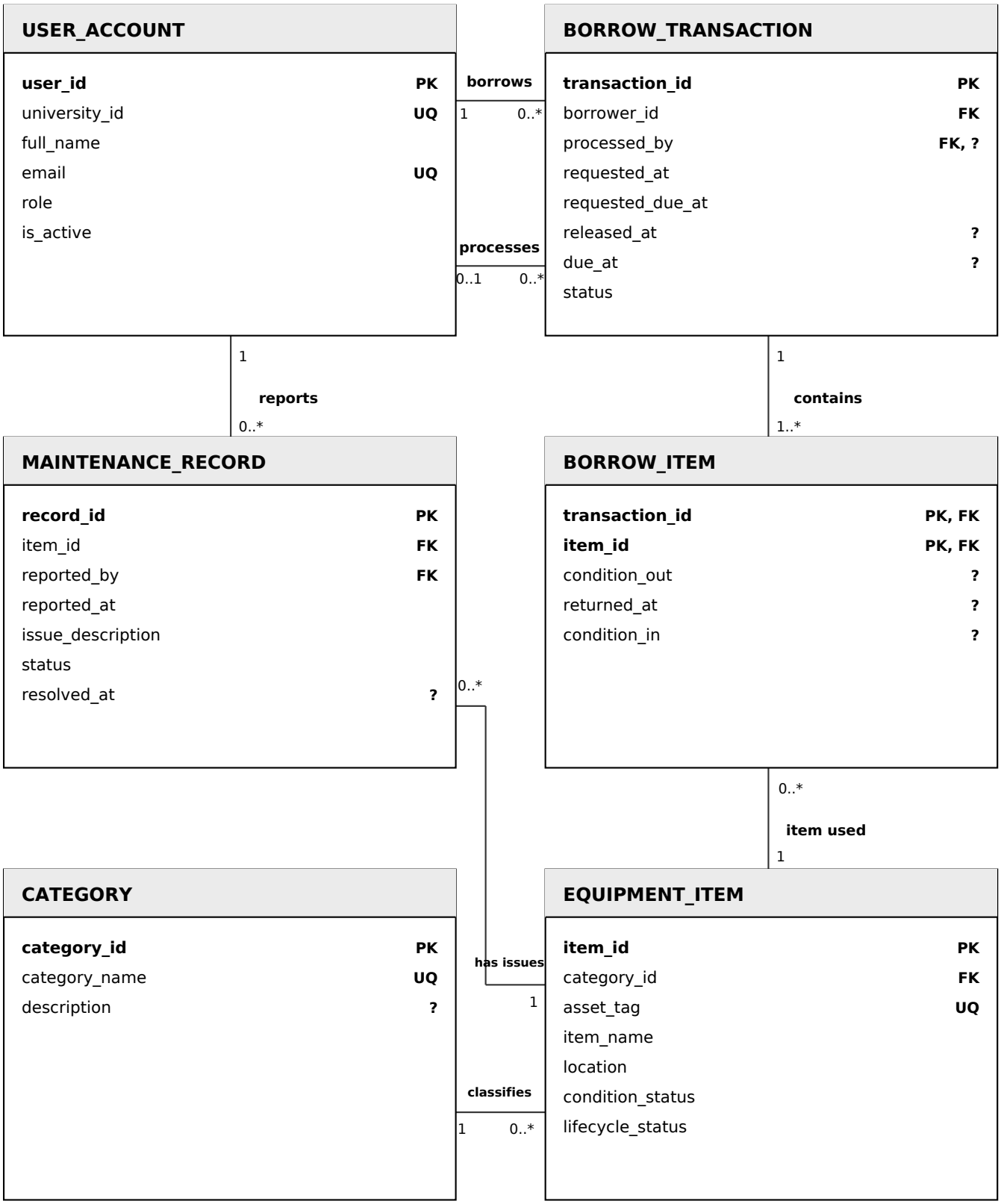
4. Database Implementation and Test Data

The accompanying .sql file creates the database and six InnoDB tables with primary keys, foreign keys, unique constraints, status restrictions, date checks, and indexes. It populates 10 users, 5 categories, 12 equipment units, 9 borrowing transactions, 15 borrowing-item rows, and 4 maintenance records.

The script ends with SHOW TABLES, SELECT * statements for every table, row-count checks, an active-loan query, and two diagnostics. Multi-row operational rules, such as preventing simultaneous releases, require transactional application checks and equipment-row locking in addition to these table constraints.

5. Updated Entity-Relationship Diagram (ERD)

Keys, referential links, cardinalities, and optional participation are shown below.



Legend: PK = primary key; FK = foreign key; UQ = unique; ? = nullable.

Cardinalities: 1 = exactly one; 0..1 = optional one; 0..* = zero or many; 1..* = one or many.

Participation: each transaction must have one borrower and, before release, at least one item; each item has one category. Request processing is optional until a staff decision is made. A transaction cannot be closed until all its borrowed items have been returned.